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SYSTEMS AND METHODS FOR DETECTING CONTACT BETWEEN A LINK AND AN EXTERNAL OBJECT

Abstract

Systems and methods for detecting contact between a link and an external object are provided. In one aspect, there is provided a robotic system, including a manipulatable link, a rigid shell configured to overlay the manipulatable link, and one or more sensors positioned between the rigid shell and the manipulatable link. The one or more sensors are configured to detect contact between the rigid shell and an external object.

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Background/Summary

RELATED APPLICATION [0001] This present application is a continuation of U.S. patent application Ser. No. 18/497,257, filed Oct. 30, 2023, which is a continuation of U.S. patent application Ser. No. 17/355,151 filed Jun. 22, 2021, which claims the benefit of, and priority to, U.S. Provisional Patent Application No. 63/045,351 filed Jun. 29, 2020, each of which are incorporated herein by reference in their entirety for all purposes.

TECHNOLOGICAL FIELD

[0002] The systems and methods disclosed herein are directed to systems and methods for robotic medical systems, and more particularly to detecting contact between a link and an external object.

BACKGROUND

[0003] Robotically-enabled medical system may be capable of performing a variety of medical procedures, including both minimally invasive, such as laparoscopy, and non-invasive, such as endoscopy, procedures. Among endoscopic procedures, the system may be capable of performing bronchoscopy, ureteroscopy, gastroscopy, etc.

[0004] Such robotic medical systems may include robotic arms configured to control the movement of medical tool(s) during a given medical procedure. In order to achieve a desired pose of a medical tool, a robotic arm may be placed into a pose that may cause the robotic arm to come into contact with another object in the environment. Unintended contact or collisions with such external objects can adversely impact the robotic medical system and/or the external object, and thus it is desirable to detect such collisions.

SUMMARY

[0005] The systems, methods and devices of this disclosure each have several innovative aspects, no single one of which is solely responsible for the desirable attributes disclosed herein.

[0006] In one aspect, there is provided a robotic system, comprising: a manipulatable link; a rigid shell configured to overlay the manipulatable link; and one or more sensors positioned between the rigid shell and the manipulatable link, the one or more sensors configured to detect contact between the rigid shell and an external object.

[0007] In certain implementations, the at least one of the one or more sensors can comprise a beam flexure.

[0008] In certain implementations, the beam flexure is configured to engage with an underlying force sensor.

[0009] In certain implementations, the manipulatable link comprises a link of a robotic arm.

[0010] In certain implementations, the manipulatable link comprises a first link or a second link of the robotic arm.

[0011] In certain implementations, at least two or more sensors are located along the proximal link or the distal link of the robotic arm for sensing contact with the respective link.

[0012] In certain implementations, the external object comprises a patient, clinician, or inanimate object.

[0013] In certain implementations, the rigid shell is suspended over the manipulatable link via the one or more sensors.

[0014] In certain implementations, the one or more sensors are further configured to detect a direction of the contact between the rigid shell and the external object.

[0015] In certain implementations, the one or more sensors are further configured to measure a

magnitude of a force resulting from the contact between the rigid shell and the external object.

[0016] In certain implementations, the one or more sensors comprise a first set of one or more sensors and a second set of one or more sensors, the first set of one or more sensor positioned between the rigid shell and the manipulatable link at a proximal end of the manipulatable link, and the second set of one or more sensor positioned between the rigid shell and the manipulatable link at a distal end of the manipulatable link.

[0017] In certain implementations, the one or more sensors are further configured to detect a direction of the contact between the rigid shell and the one or more external objects.

[0018] In certain implementations, the one or more sensors are further configured to measure a magnitude of a force resulting from the contact between the rigid shell and the one or more external objects.

[0019] In certain implementations, the one or more sensors are further configured to measure a torque resulting from the contact between the rigid shell and the one or more external objects.

[0020] In certain implementations, the one or more sensors are further configured to measure a direction of relative motion between the rigid shell and the link.

[0021] In another aspect, there is provided a sensor assembly for detecting contact with a manipulatable link of a robotic system, the sensor assembly comprising: a beam flexure; an underlying force sensor for detecting contact with the manipulatable link; and a circuit board.

[0022] In certain implementations, the beam flexure comprises a first end that is fixed and a second end coupled to the force sensor.

[0023] In certain implementations, the beam flexure is engaged with a rigid shell at a point between the first end and the second end.

[0024] In certain implementations, the force exerted on the sensor is proportional to the force applied to the point at which the shell engages the flexure.

[0025] In certain implementations, the beam flexure is engaged with a rigid shell at a fixed point along the beam flexure.

[0026] In certain implementations, a processor is configured to receive an output from the force sensor and generate a force value indicative of a force applied to the rigid shell.

[0027] In yet another aspect, there is provided a robotic system, comprising: a manipulatable link; a rigid shell configured to overlay the manipulatable link; and one or more sensors positioned between the rigid shell and the manipulatable link, the one or more sensors configured to detect contact between one or more external objects and the rigid shell in at least two or more directions.

[0028] In certain implementations, the one or more sensors are further configured to detect contact between the one or more external objects and the rigid shell in one or more rotational directions.

[0029] In certain implementations, at least one of the one or more sensors comprises a beam flexure.

[0030] In certain implementations, the beam flexure engages with an underlying micro-force sensor

[0031] In certain implementations, the manipulatable link comprises a link of a robotic arm.

[0032] In certain implementations, the manipulatable link comprises a proximal link or a distal link of the robotic arm.

[0033] In certain implementations, at least two or more sensors are located along the proximal link or the distal link of the robotic arm for sensing contact with the respective link.

[0034] In certain implementations, the external object comprises a patient, staff or, inanimate object.

[0035] In still yet another aspect, there is provided a method of detecting contact between a robotic manipulator and an external object, the method comprising: receiving a signal from one or more sensors positioned between a link of the robotic manipulator and a rigid shell configured to overlay the link, the received signal indicative of the degree of contact between the rigid shell and the one or more sensors; and determining that contact between the robotic manipulator and the external object has occurred in response to the received signal meeting or exceeding a threshold value.

[0036] In certain implementations, the method further comprises: detecting a direction of the contact between the rigid shell and the external object based on the signal received from the one or more sensors.

[0037] In certain implementations, the method further comprises: measuring a magnitude of a force resulting from the contact between the rigid shell and the external object based on the signal received from the one or more sensors.

[0038] In certain implementations, the method further comprises: measuring a torque resulting from the contact between the rigid shell and the external object based on the signal received from the one or more sensors.

[0039] In certain implementations, the method further comprises: measuring a direction of relative motion between the rigid shell and the link.

[0040] In another aspect, there is provided a robotic arm, comprising: a link; a rigid shell formed over the link; and one or more sensors configured to detect contact between an external object and the rigid shell.

[0041] In certain implementations, the external object comprises one or more of: a patient, a clinician, or an inanimate object.

[0042] In certain implementations, the one or more sensors are positioned in an interstitial space between the link and the rigid shell.

[0043] In certain implementations, the one or more sensors are further configured to measure a magnitude of a force applied to the link.

[0044] In certain implementations, the one or more sensors are further configured to measure a direction of a force applied to the link.

[0045] In certain implementations, the one or more sensors are further configured to measure a torque applied to the link.

[0046] In certain implementations, the one or more sensors comprise a plurality of sensors positioned at different locations of the link.

[0047] In certain implementations, the one or more sensors comprise a plurality of circuit board mounted force sensors.

[0048] In certain implementations, the one or more sensors comprise a plurality of strain gauges.

[0049] In certain implementations, the one or more sensors comprise a plurality of force sensing resistors.

[0050] In certain implementations, the one or more sensors comprise a plurality of capacitive sensors.

[0051] In certain implementations, the one or more sensors comprise a plurality of sensors that support the rigid shell without being fixed to the rigid shell.

[0052] In certain implementations, a given sensor of the one or more sensors comprises a load concentrator configured to control a point at which the rigid shell applies a force to the given sensor.

[0053] In certain implementations, a given sensor of the one or more sensors comprises a flexure configured to scale a force applied to the given sensor and isolate the given sensor from shear forces.

[0054] In certain implementations, the one or more sensors are further configured to measure a relative position of the rigid shell relative to the link.

[0055] In certain implementations, the robotic arm further comprises: a support configured to support the rigid shell relative to the link.

[0056] In certain implementations, the support comprises one or more of: a spring, a flexure, or a suspension.

[0057] In certain implementations, the one or more sensors are further configured to measure a direction of relative motion between the rigid shell and the link.

[0058] In certain implementations, the one or more sensors are further configured to determine a direction of a force applied to the rigid shell.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0059] The disclosed aspects will hereinafter be described in conjunction with the appended drawings, provided to illustrate and not to limit the disclosed aspects, wherein like designations denote like elements.

[0060] FIG. 1 illustrates an embodiment of a cart-based robotic system arranged for diagnostic and/or therapeutic bronchoscopy.

[0061] FIG. 2 depicts further aspects of the robotic system of FIG. 1.

[0062] FIG. 3 illustrates an embodiment of the robotic system of FIG. 1 arranged for ureteroscopy.

[0063] FIG. 4 illustrates an embodiment of the robotic system of FIG. 1 arranged for a vascular procedure.

[0064] FIG. 5 illustrates an embodiment of a table-based robotic system arranged for a bronchoscopic procedure.

[0065] FIG. 6 provides an alternative view of the robotic system of FIG. 5.

[0066] FIG. 7 illustrates an example system configured to stow robotic arm(s).

[0067] FIG. 8 illustrates an embodiment of a table-based robotic system configured for a ureteroscopic procedure.

[0068] FIG. 9 illustrates an embodiment of a table-based robotic system configured for a laparoscopic procedure.

[0069] FIG. 10 illustrates an embodiment of the table-based robotic system of FIGS. 5-9 with pitch or tilt adjustment.

[0070] FIG. 11 provides a detailed illustration of the interface between the table and the column of the table-based robotic system of FIGS. 5-10.

[0071] FIG. 12 illustrates an alternative embodiment of a table-based robotic system.

[0072] FIG. 13 illustrates an end view of the table-based robotic system of FIG. 12.

[0073] FIG. 14 illustrates an end view of a table-based robotic system with robotic arms attached thereto.

[0074] FIG. 15 illustrates an exemplary instrument driver.

[0075] FIG. 16 illustrates an exemplary medical instrument with a paired instrument driver.

[0076] FIG. 17 illustrates an alternative design for an instrument driver and instrument where the axes of the drive units are parallel to the axis of the elongated shaft of the instrument.

[0077] FIG. 18 illustrates an instrument having an instrument-based insertion architecture.

[0078] FIG. 19 illustrates an exemplary controller, according to some embodiments.

[0079] FIG. 20 depicts a block diagram illustrating a localization system that estimates a location of one or more elements of the robotic systems of FIGS. 1-10, such as the location of the instrument of FIGS. 16-18, in accordance to an example embodiment.

[0080] FIGS. 21A and 21B illustrate example robotic links that can form a portion of a robotic arm, in accordance with aspects of this disclosure.

[0081] FIG. 22 illustrates an example link including a rigid shell and configured to detect contact with external objects, in accordance with aspects of this disclosure.

[0082] FIG. 23 illustrates one example of a force-sensing connection that can be used in the link of FIG. 22, in accordance with aspects of this disclosure.

[0083] FIG. 24A illustrates two views of an embodiment in which fourteen sensors are included in a link, in accordance with aspects of this disclosure.

[0084] FIG. 24B illustrates two view of an embodiment in which twelve sensors are included in a

link, in accordance with aspects of this disclosure.

[0085] FIG. **25** is a flowchart illustrating an example method operable by a surgical robotic system, or component(s) thereof, for detecting contact between a robotic manipulator and an external object, in accordance with aspects of this disclosure.

[0086] FIG. **26** illustrates an embodiment of a beam flexure sensor, in accordance with aspects of this disclosure.

[0087] FIG. **27** illustrates another embodiment of a beam flexure sensor, in accordance with aspects of this disclosure.

[0088] FIG. **28** illustrates an example robotic link that can form a portion of a robotic arm in accordance with aspects of this disclosure.

DETAILED DESCRIPTION

1. Overview

[0089] Aspects of the present disclosure may be integrated into a robotically-enabled medical system capable of performing a variety of medical procedures, including both minimally invasive, such as laparoscopy, and non-invasive, such as endoscopy, procedures. Among endoscopic procedures, the system may be capable of performing bronchoscopy, ureteroscopy, gastroscopy, etc.

[0090] In addition to performing the breadth of procedures, the system may provide additional benefits, such as enhanced imaging and guidance to assist the physician. Additionally, the system may provide the physician with the ability to perform the procedure from an ergonomic position without the need for awkward arm motions and positions. Still further, the system may provide the physician with the ability to perform the procedure with improved ease of use such that one or more of the instruments of the system can be controlled by a single user.

[0091] Various embodiments will be described below in conjunction with the drawings for purposes of illustration. It should be appreciated that many other implementations of the disclosed concepts are possible, and various advantages can be achieved with the disclosed implementations. Headings are included herein for reference and to aid in locating various sections. These headings are not intended to limit the scope of the concepts described with respect thereto. Such concepts may have applicability throughout the entire specification.

A. Robotic System—Cart.

[0092] The robotically-enabled medical system may be configured in a variety of ways depending on the particular procedure. FIG. **1** illustrates an embodiment of a cart-based robotically-enabled system **10** arranged for a diagnostic and/or therapeutic bronchoscopy. During a bronchoscopy, the system **10** may comprise a cart **11** having one or more robotic arms **12** to deliver a medical instrument, such as a steerable endoscope **13**, which may be a procedure-specific bronchoscope for bronchoscopy, to a natural orifice access point (i.e., the mouth of the patient positioned on a table in the present example) to deliver diagnostic and/or therapeutic tools. As shown, the cart **11** may be positioned proximate to the patient's upper torso in order to provide access to the access point. Similarly, the robotic arms **12** may be actuated to position the bronchoscope relative to the access point. The arrangement in FIG. **1** may also be utilized when performing a gastro-intestinal (GI) procedure with a gastroscope, a specialized endoscope for GI procedures. FIG. **2** depicts an example embodiment of the cart in greater detail.

[0093] With continued reference to FIG. **1**, once the cart **11** is properly positioned, the robotic arms **12** may insert the steerable endoscope **13** into the patient robotically, manually, or a combination thereof. As shown, the steerable endoscope **13** may comprise at least two telescoping parts, such as an inner leader portion and an outer sheath portion, each portion coupled to a separate instrument driver from the set of instrument drivers **28**, each instrument driver coupled to the distal end of an individual robotic arm. This linear arrangement of the instrument drivers **28**, which facilitates coaxially aligning the leader portion with the sheath portion, creates a “virtual rail” **29** that may be repositioned in space by manipulating the one or more robotic arms **12** into different angles and/or

positions. The virtual rails described herein are depicted in the Figures using dashed lines, and accordingly the dashed lines do not depict any physical structure of the system. Translation of the instrument drivers **28** along the virtual rail **29** telescopes the inner leader portion relative to the outer sheath portion or advances or retracts the endoscope **13** from the patient. The angle of the virtual rail **29** may be adjusted, translated, and pivoted based on clinical application or physician preference. For example, in bronchoscopy, the angle and position of the virtual rail **29** as shown represents a compromise between providing physician access to the endoscope **13** while minimizing friction that results from bending the endoscope **13** into the patient's mouth.

[0094] The endoscope **13** may be directed down the patient's trachea and lungs after insertion using precise commands from the robotic system until reaching the target destination or operative site. In order to enhance navigation through the patient's lung network and/or reach the desired target, the endoscope **13** may be manipulated to telescopically extend the inner leader portion from the outer sheath portion to obtain enhanced articulation and greater bend radius. The use of separate instrument drivers **28** also allows the leader portion and sheath portion to be driven independently of each other.

[0095] For example, the endoscope **13** may be directed to deliver a biopsy needle to a target, such as, for example, a lesion or nodule within the lungs of a patient. The needle may be deployed down a working channel that runs the length of the endoscope to obtain a tissue sample to be analyzed by a pathologist. Depending on the pathology results, additional tools may be deployed down the working channel of the endoscope for additional biopsies. After identifying a nodule to be malignant, the endoscope **13** may endoscopically deliver tools to resect the potentially cancerous tissue. In some instances, diagnostic and therapeutic treatments can be delivered in separate procedures. In those circumstances, the endoscope **13** may also be used to deliver a fiducial to "mark" the location of the target nodule as well. In other instances, diagnostic and therapeutic treatments may be delivered during the same procedure.

[0096] The system **10** may also include a movable tower **30**, which may be connected via support cables to the cart **11** to provide support for controls, electronics, fluidics, optics, sensors, and/or power to the cart **11**. Placing such functionality in the tower **30** allows for a smaller form factor cart **11** that may be more easily adjusted and/or re-positioned by an operating physician and his/her staff. Additionally, the division of functionality between the cart/table and the support tower **30** reduces operating room clutter and facilitates improving clinical workflow. While the cart **11** may be positioned close to the patient, the tower **30** may be stowed in a remote location to stay out of the way during a procedure.

[0097] In support of the robotic systems described above, the tower **30** may include component(s) of a computer-based control system that stores computer program instructions, for example, within a non-transitory computer-readable storage medium such as a persistent magnetic storage drive, solid state drive, etc. The execution of those instructions, whether the execution occurs in the tower **30** or the cart **11**, may control the entire system or sub-system(s) thereof. For example, when executed by a processor of the computer system, the instructions may cause the components of the robotics system to actuate the relevant carriages and arm mounts, actuate the robotics arms, and control the medical instruments. For example, in response to receiving the control signal, the motors in the joints of the robotics arms may position the arms into a certain posture.

[0098] The tower **30** may also include a pump, flow meter, valve control, and/or fluid access in order to provide controlled irrigation and aspiration capabilities to the system that may be deployed through the endoscope **13**. These components may also be controlled using the computer system of the tower **30**. In some embodiments, irrigation and aspiration capabilities may be delivered directly to the endoscope **13** through separate cable(s).

[0099] The tower **30** may include a voltage and surge protector designed to provide filtered and protected electrical power to the cart **11**, thereby avoiding placement of a power transformer and other auxiliary power components in the cart **11**, resulting in a smaller, more moveable cart **11**.

[0100] The tower **30** may also include support equipment for the sensors deployed throughout the robotic system **10**. For example, the tower **30** may include optoelectronics equipment for detecting, receiving, and processing data received from the optical sensors or cameras throughout the robotic system **10**. In combination with the control system, such optoelectronics equipment may be used to generate real-time images for display in any number of consoles deployed throughout the system, including in the tower **30**. Similarly, the tower **30** may also include an electronic subsystem for receiving and processing signals received from deployed electromagnetic (EM) sensors. The tower **30** may also be used to house and position an EM field generator for detection by EM sensors in or on the medical instrument.

[0101] The tower **30** may also include a console **31** in addition to other consoles available in the rest of the system, e.g., console mounted on top of the cart. The console **31** may include a user interface and a display screen, such as a touchscreen, for the physician operator. Consoles in the system **10** are generally designed to provide both robotic controls as well as preoperative and real-time information of the procedure, such as navigational and localization information of the endoscope **13**. When the console **31** is not the only console available to the physician, it may be used by a second operator, such as a nurse, to monitor the health or vitals of the patient and the operation of the system **10**, as well as to provide procedure-specific data, such as navigational and localization information. In other embodiments, the console **31** is housed in a body that is separate from the tower **30**.

[0102] The tower **30** may be coupled to the cart **11** and endoscope **13** through one or more cables or connections (not shown). In some embodiments, the support functionality from the tower **30** may be provided through a single cable to the cart **11**, simplifying and de-cluttering the operating room. In other embodiments, specific functionality may be coupled in separate cabling and connections. For example, while power may be provided through a single power cable to the cart **11**, the support for controls, optics, fluidics, and/or navigation may be provided through a separate cable.

[0103] FIG. 2 provides a detailed illustration of an embodiment of the cart **11** from the cart-based robotically-enabled system shown in FIG. 1. The cart **11** generally includes an elongated support structure **14** (often referred to as a “column”), a cart base **15**, and a console **16** at the top of the column **14**. The column **14** may include one or more carriages, such as a carriage **17** (alternatively “arm support”) for supporting the deployment of one or more robotic arms **12** (three shown in FIG. 2). The carriage **17** may include individually configurable arm mounts that rotate along a perpendicular axis to adjust the base of the robotic arms **12** for better positioning relative to the patient. The carriage **17** also includes a carriage interface **19** that allows the carriage **17** to vertically translate along the column **14**.

[0104] The carriage interface **19** is connected to the column **14** through slots, such as slot **20**, that are positioned on opposite sides of the column **14** to guide the vertical translation of the carriage **17**. The slot **20** contains a vertical translation interface to position and hold the carriage **17** at various vertical heights relative to the cart base **15**. Vertical translation of the carriage **17** allows the cart **11** to adjust the reach of the robotic arms **12** to meet a variety of table heights, patient sizes, and physician preferences. Similarly, the individually configurable arm mounts on the carriage **17** allow the robotic arm base **21** of the robotic arms **12** to be angled in a variety of configurations.

[0105] In some embodiments, the slot **20** may be supplemented with slot covers that are flush and parallel to the slot surface to prevent dirt and fluid ingress into the internal chambers of the column **14** and the vertical translation interface as the carriage **17** vertically translates. The slot covers may be deployed through pairs of spring spools positioned near the vertical top and bottom of the slot **20**. The covers are coiled within the spools until deployed to extend and retract from their coiled state as the carriage **17** vertically translates up and down. The spring-loading of the spools provides force to retract the cover into a spool when the carriage **17** translates towards the spool, while also maintaining a tight seal when the carriage **17** translates away from the spool. The covers may be

connected to the carriage **17** using, for example, brackets in the carriage interface **19** to facilitate proper extension and retraction of the cover as the carriage **17** translates.

[0106] The column **14** may internally comprise mechanisms, such as gears and motors, that are designed to use a vertically aligned lead screw to translate the carriage **17** in a mechanized fashion in response to control signals generated in response to user inputs, e.g., inputs from the console **16**.

[0107] The robotic arms **12** may generally comprise robotic arm bases **21** and end effectors **22**, separated by a series of links **23** that are connected by a series of joints **24**, each joint comprising an independent actuator, each actuator comprising an independently controllable motor. Each independently controllable joint represents an independent degree of freedom available to the robotic arm **12**. Each of the robotic arms **12** may have seven joints, and thus provide seven degrees of freedom. A multitude of joints result in a multitude of degrees of freedom, allowing for “redundant” degrees of freedom. Having redundant degrees of freedom allows the robotic arms **12** to position their respective end effectors **22** at a specific position, orientation, and trajectory in space using different link positions and joint angles. This allows for the system to position and direct a medical instrument from a desired point in space while allowing the physician to move the arm joints into a clinically advantageous position away from the patient to create greater access, while avoiding arm collisions.

[0108] The cart base **15** balances the weight of the column **14**, carriage **17**, and robotic arms **12** over the floor. Accordingly, the cart base **15** houses heavier components, such as electronics, motors, power supply, as well as components that either enable movement and/or immobilize the cart **11**. For example, the cart base **15** includes rollable wheel-shaped casters **25** that allow for the cart **11** to easily move around the room prior to a procedure. After reaching the appropriate position, the casters **25** may be immobilized using wheel locks to hold the cart **11** in place during the procedure.

[0109] Positioned at the vertical end of the column **14**, the console **16** allows for both a user interface for receiving user input and a display screen (or a dual-purpose device such as, for example, a touchscreen **26**) to provide the physician user with both preoperative and intraoperative data. Potential preoperative data on the touchscreen **26** may include preoperative plans, navigation and mapping data derived from preoperative computerized tomography (CT) scans, and/or notes from preoperative patient interviews. Intraoperative data on display may include optical information provided from the tool, sensor and coordinate information from sensors, as well as vital patient statistics, such as respiration, heart rate, and/or pulse. The console **16** may be positioned and tilted to allow a physician to access the console **16** from the side of the column **14** opposite the carriage **17**. From this position, the physician may view the console **16**, robotic arms **12**, and patient while operating the console **16** from behind the cart **11**. As shown, the console **16** also includes a handle **27** to assist with maneuvering and stabilizing the cart **11**.

[0110] FIG. 3 illustrates an embodiment of a robotically-enabled system **10** arranged for ureteroscopy. In a ureteroscopic procedure, the cart **11** may be positioned to deliver a ureteroscope **32**, a procedure-specific endoscope designed to traverse a patient's urethra and ureter, to the lower abdominal area of the patient. In a ureteroscopy, it may be desirable for the ureteroscope **32** to be directly aligned with the patient's urethra to reduce friction and forces on the sensitive anatomy in the area. As shown, the cart **11** may be aligned at the foot of the table to allow the robotic arms **12** to position the ureteroscope **32** for direct linear access to the patient's urethra. From the foot of the table, the robotic arms **12** may insert the ureteroscope **32** along the virtual rail **33** directly into the patient's lower abdomen through the urethra.

[0111] After insertion into the urethra, using similar control techniques as in bronchoscopy, the ureteroscope **32** may be navigated into the bladder, ureters, and/or kidneys for diagnostic and/or therapeutic applications. For example, the ureteroscope **32** may be directed into the ureter and kidneys to break up kidney stone build up using a laser or ultrasonic lithotripsy device deployed down the working channel of the ureteroscope **32**. After lithotripsy is complete, the resulting stone

fragments may be removed using baskets deployed down the ureteroscope **32**.

[0112] FIG. **4** illustrates an embodiment of a robotically-enabled system **10** similarly arranged for a vascular procedure. In a vascular procedure, the system **10** may be configured such that the cart **11** may deliver a medical instrument **34**, such as a steerable catheter, to an access point in the femoral artery in the patient's leg. The femoral artery presents both a larger diameter for navigation as well as a relatively less circuitous and tortuous path to the patient's heart, which simplifies navigation. As in a ureteroscopic procedure, the cart **11** may be positioned towards the patient's legs and lower abdomen to allow the robotic arms **12** to provide a virtual rail **35** with direct linear access to the femoral artery access point in the patient's thigh/hip region. After insertion into the artery, the medical instrument **34** may be directed and inserted by translating the instrument drivers **28**. Alternatively, the cart may be positioned around the patient's upper abdomen in order to reach alternative vascular access points, such as, for example, the carotid and brachial arteries near the shoulder and wrist.

B. Robotic System—Table.

[0113] Embodiments of the robotically-enabled medical system may also incorporate the patient's table. Incorporation of the table reduces the amount of capital equipment within the operating room by removing the cart, which allows greater access to the patient. FIG. **5** illustrates an embodiment of such a robotically-enabled system arranged for a bronchoscopic procedure. System **36** includes a support structure or column **37** for supporting platform **38** (shown as a “table” or “bed”) over the floor. Much like in the cart-based systems, the end effectors of the robotic arms **39** of the system **36** comprise instrument drivers **42** that are designed to manipulate an elongated medical instrument, such as a bronchoscope **40** in FIG. **5**, through or along a virtual rail **41** formed from the linear alignment of the instrument drivers **42**. In practice, a C-arm for providing fluoroscopic imaging may be positioned over the patient's upper abdominal area by placing the emitter and detector around the table **38**.

[0114] FIG. **6** provides an alternative view of the system **36** without the patient and medical instrument for discussion purposes. As shown, the column **37** may include one or more carriages **43** shown as ring-shaped in the system **36**, from which the one or more robotic arms **39** may be based. The carriages **43** may translate along a vertical column interface **44** that runs the length of the column **37** to provide different vantage points from which the robotic arms **39** may be positioned to reach the patient. The carriage(s) **43** may rotate around the column **37** using a mechanical motor positioned within the column **37** to allow the robotic arms **39** to have access to multiples sides of the table **38**, such as, for example, both sides of the patient. In embodiments with multiple carriages, the carriages may be individually positioned on the column and may translate and/or rotate independently of the other carriages. While the carriages **43** need not surround the column **37** or even be circular, the ring-shape as shown facilitates rotation of the carriages **43** around the column **37** while maintaining structural balance. Rotation and translation of the carriages **43** allows the system **36** to align the medical instruments, such as endoscopes and laparoscopes, into different access points on the patient. In other embodiments (not shown), the system **36** can include a patient table or bed with adjustable arm supports in the form of bars or rails extending alongside it. One or more robotic arms **39** (e.g., via a shoulder with an elbow joint) can be attached to the adjustable arm supports, which can be vertically adjusted. By providing vertical adjustment, the robotic arms **39** are advantageously capable of being stowed compactly beneath the patient table or bed, and subsequently raised during a procedure.

[0115] The robotic arms **39** may be mounted on the carriages **43** through a set of arm mounts **45** comprising a series of joints that may individually rotate and/or telescopically extend to provide additional configurability to the robotic arms **39**. Additionally, the arm mounts **45** may be positioned on the carriages **43** such that, when the carriages **43** are appropriately rotated, the arm mounts **45** may be positioned on either the same side of the table **38** (as shown in FIG. **6**), on opposite sides of the table **38** (as shown in FIG. **9**), or on adjacent sides of the table **38** (not shown).

[0116] The column **37** structurally provides support for the table **38**, and a path for vertical translation of the carriages **43**. Internally, the column **37** may be equipped with lead screws for guiding vertical translation of the carriages, and motors to mechanize the translation of the carriages **43** based the lead screws. The column **37** may also convey power and control signals to the carriages **43** and the robotic arms **39** mounted thereon.

[0117] The table base **46** serves a similar function as the cart base **15** in the cart **11** shown in FIG. **2**, housing heavier components to balance the table/bed **38**, the column **37**, the carriages **43**, and the robotic arms **39**. The table base **46** may also incorporate rigid casters to provide stability during procedures. Deployed from the bottom of the table base **46**, the casters may extend in opposite directions on both sides of the base **46** and retract when the system **36** needs to be moved.

[0118] With continued reference to FIG. **6**, the system **36** may also include a tower (not shown) that divides the functionality of the system **36** between the table and the tower to reduce the form factor and bulk of the table. As in earlier disclosed embodiments, the tower may provide a variety of support functionalities to the table, such as processing, computing, and control capabilities, power, fluidics, and/or optical and sensor processing. The tower may also be movable to be positioned away from the patient to improve physician access and de-clutter the operating room. Additionally, placing components in the tower allows for more storage space in the table base **46** for potential stowage of the robotic arms **39**. The tower may also include a master controller or console that provides both a user interface for user input, such as keyboard and/or pendant, as well as a display screen (or touchscreen) for preoperative and intraoperative information, such as real-time imaging, navigation, and tracking information. In some embodiments, the tower may also contain holders for gas tanks to be used for insufflation.

[0119] In some embodiments, a table base may stow and store the robotic arms when not in use. FIG. **7** illustrates a system **47** that stows robotic arms in an embodiment of the table-based system. In the system **47**, carriages **48** may be vertically translated into base **49** to stow robotic arms **50**, arm mounts **51**, and the carriages **48** within the base **49**. Base covers **52** may be translated and retracted open to deploy the carriages **48**, arm mounts **51**, and robotic arms **50** around column **53**, and closed to stow to protect them when not in use. The base covers **52** may be sealed with a membrane **54** along the edges of its opening to prevent dirt and fluid ingress when closed.

[0120] FIG. **8** illustrates an embodiment of a robotically-enabled table-based system configured for a ureteroscopy procedure. In a ureteroscopy, the table **38** may include a swivel portion **55** for positioning a patient off-angle from the column **37** and table base **46**. The swivel portion **55** may rotate or pivot around a pivot point (e.g., located below the patient's head) in order to position the bottom portion of the swivel portion **55** away from the column **37**. For example, the pivoting of the swivel portion **55** allows a C-arm (not shown) to be positioned over the patient's lower abdomen without competing for space with the column (not shown) below table **38**. By rotating the carriage (not shown) around the column **37**, the robotic arms **39** may directly insert a ureteroscope **56** along a virtual rail **57** into the patient's groin area to reach the urethra. In a ureteroscopy, stirrups **58** may also be fixed to the swivel portion **55** of the table **38** to support the position of the patient's legs during the procedure and allow clear access to the patient's groin area.

[0121] In a laparoscopic procedure, through small incision(s) in the patient's abdominal wall, minimally invasive instruments may be inserted into the patient's anatomy. In some embodiments, the minimally invasive instruments comprise an elongated rigid member, such as a shaft, which is used to access anatomy within the patient. After inflation of the patient's abdominal cavity, the instruments may be directed to perform surgical or medical tasks, such as grasping, cutting, ablating, suturing, etc. In some embodiments, the instruments can comprise a scope, such as a laparoscope. FIG. **9** illustrates an embodiment of a robotically-enabled table-based system configured for a laparoscopic procedure. As shown in FIG. **9**, the carriages **43** of the system **36** may be rotated and vertically adjusted to position pairs of the robotic arms **39** on opposite sides of the table **38**, such that instrument **59** may be positioned using the arm mounts **45** to be passed

through minimal incisions on both sides of the patient to reach his/her abdominal cavity.

[0122] To accommodate laparoscopic procedures, the robotically-enabled table system may also tilt the platform to a desired angle. FIG. 10 illustrates an embodiment of the robotically-enabled medical system with pitch or tilt adjustment. As shown in FIG. 10, the system 36 may accommodate tilt of the table 38 to position one portion of the table at a greater distance from the floor than the other. Additionally, the arm mounts 45 may rotate to match the tilt such that the robotic arms 39 maintain the same planar relationship with the table 38. To accommodate steeper angles, the column 37 may also include telescoping portions 60 that allow vertical extension of the column 37 to keep the table 38 from touching the floor or colliding with the table base 46.

[0123] FIG. 11 provides a detailed illustration of the interface between the table 38 and the column 37. Pitch rotation mechanism 61 may be configured to alter the pitch angle of the table 38 relative to the column 37 in multiple degrees of freedom. The pitch rotation mechanism 61 may be enabled by the positioning of orthogonal axes 1, 2 at the column-table interface, each axis actuated by a separate motor 3, 4 responsive to an electrical pitch angle command. Rotation along one screw 5 would enable tilt adjustments in one axis 1, while rotation along the other screw 6 would enable tilt adjustments along the other axis 2. In some embodiments, a ball joint can be used to alter the pitch angle of the table 38 relative to the column 37 in multiple degrees of freedom.

[0124] For example, pitch adjustments are particularly useful when trying to position the table in a Trendelenburg position, i.e., position the patient's lower abdomen at a higher position from the floor than the patient's upper abdomen, for lower abdominal surgery. The Trendelenburg position causes the patient's internal organs to slide towards his/her upper abdomen through the force of gravity, clearing out the abdominal cavity for minimally invasive tools to enter and perform lower abdominal surgical or medical procedures, such as laparoscopic prostatectomy.

[0125] FIGS. 12 and 13 illustrate isometric and end views of an alternative embodiment of a table-based surgical robotics system 100. The surgical robotics system 100 includes one or more adjustable arm supports 105 that can be configured to support one or more robotic arms (see, for example, FIG. 14) relative to a table 101. In the illustrated embodiment, a single adjustable arm support 105 is shown, though an additional arm support 105 can be provided on an opposite side of the table 101. The adjustable arm support 105 can be configured so that it can move relative to the table 101 to adjust and/or vary the position of the adjustable arm support 105 and/or any robotic arms mounted thereto relative to the table 101. For example, the adjustable arm support 105 may be adjusted one or more degrees of freedom relative to the table 101. The adjustable arm support 105 provides high versatility to the system 100, including the ability to easily stow the one or more adjustable arm supports 105 and any robotics arms attached thereto beneath the table 101. The adjustable arm support 105 can be elevated from the stowed position to a position below an upper surface of the table 101. In other embodiments, the adjustable arm support 105 can be elevated from the stowed position to a position above an upper surface of the table 101.

[0126] The adjustable arm support 105 can provide several degrees of freedom, including lift, lateral translation, tilt, etc. In the illustrated embodiment of FIGS. 12 and 13, the arm support 105 is configured with four degrees of freedom, which are illustrated with arrows in FIG. 12. A first degree of freedom allows for adjustment of the adjustable arm support 105 in the z-direction ("Z-lift"). For example, the adjustable arm support 105 can include a carriage 109 configured to move up or down along or relative to a column 102 supporting the table 101. A second degree of freedom can allow the adjustable arm support 105 to tilt. For example, the adjustable arm support 105 can include a rotary joint, which can allow the adjustable arm support 105 to be aligned with the bed in a Trendelenburg position. A third degree of freedom can allow the adjustable arm support 105 to "pivot up," which can be used to adjust a distance between a side of the table 101 and the adjustable arm support 105. A fourth degree of freedom can permit translation of the adjustable arm support 105 along a longitudinal length of the table.

[0127] The surgical robotics system 100 in FIGS. 12 and 13 can comprise a table supported by a

column **102** that is mounted to a base **103**. The base **103** and the column **102** support the table **101** relative to a support surface. A floor axis **131** and a support axis **133** are shown in FIG. **13**. [0128] The adjustable arm support **105** can be mounted to the column **102**. In other embodiments, the arm support **105** can be mounted to the table **101** or base **103**. The adjustable arm support **105** can include a carriage **109**, a bar or rail connector **111** and a bar or rail **107**. In some embodiments, one or more robotic arms mounted to the rail **107** can translate and move relative to one another. [0129] The carriage **109** can be attached to the column **102** by a first joint **113**, which allows the carriage **109** to move relative to the column **102** (e.g., such as up and down a first or vertical axis **123**). The first joint **113** can provide the first degree of freedom (Z-lift) to the adjustable arm support **105**. The adjustable arm support **105** can include a second joint **115**, which provides the second degree of freedom (tilt) for the adjustable arm support **105**. The adjustable arm support **105** can include a third joint **117**, which can provide the third degree of freedom (“pivot up”) for the adjustable arm support **105**. An additional joint **119** (shown in FIG. **13**) can be provided that mechanically constrains the third joint **117** to maintain an orientation of the rail **107** as the rail connector **111** is rotated about a third axis **127**. The adjustable arm support **105** can include a fourth joint **121**, which can provide a fourth degree of freedom (translation) for the adjustable arm support **105** along a fourth axis **129**.

[0130] FIG. **14** illustrates an end view of the surgical robotics system **140A** with two adjustable arm supports **105A**, **105B** mounted on opposite sides of a table **101**. A first robotic arm **142A** is attached to the bar or rail **107A** of the first adjustable arm support **105B**. The first robotic arm **142A** includes a base **144A** attached to the rail **107A**. The distal end of the first robotic arm **142A** includes an instrument drive mechanism **146A** that can attach to one or more robotic medical instruments or tools. Similarly, the second robotic arm **142B** includes a base **144B** attached to the rail **107B**. The distal end of the second robotic arm **142B** includes an instrument drive mechanism **146B**. The instrument drive mechanism **146B** can be configured to attach to one or more robotic medical instruments or tools.

[0131] In some embodiments, one or more of the robotic arms **142A**, **142B** comprises an arm with seven or more degrees of freedom. In some embodiments, one or more of the robotic arms **142A**, **142B** can include eight degrees of freedom, including an insertion axis (1-degree of freedom including insertion), a wrist (3-degrees of freedom including wrist pitch, yaw and roll), an elbow (1-degree of freedom including elbow pitch), a shoulder (2-degrees of freedom including shoulder pitch and yaw), and base **144A**, **144B** (1-degree of freedom including translation). In some embodiments, the insertion degree of freedom can be provided by the robotic arm **142A**, **142B**, while in other embodiments, the instrument itself provides insertion via an instrument-based insertion architecture.

C. Instrument Driver & Interface.

[0132] The end effectors of the system's robotic arms may comprise (i) an instrument driver (alternatively referred to as “instrument drive mechanism” or “instrument device manipulator”) that incorporates electro-mechanical means for actuating the medical instrument and (ii) a removable or detachable medical instrument, which may be devoid of any electro-mechanical components, such as motors. This dichotomy may be driven by the need to sterilize medical instruments used in medical procedures, and the inability to adequately sterilize expensive capital equipment due to their intricate mechanical assemblies and sensitive electronics. Accordingly, the medical instruments may be designed to be detached, removed, and interchanged from the instrument driver (and thus the system) for individual sterilization or disposal by the physician or the physician's staff. In contrast, the instrument drivers need not be changed or sterilized, and may be draped for protection.

[0133] FIG. **15** illustrates an example instrument driver. Positioned at the distal end of a robotic arm, instrument driver **62** comprises one or more drive units **63** arranged with parallel axes to provide controlled torque to a medical instrument via drive shafts **64**. Each drive unit **63** comprises

an individual drive shaft **64** for interacting with the instrument, a gear head **65** for converting the motor shaft rotation to a desired torque, a motor **66** for generating the drive torque, an encoder **67** to measure the speed of the motor shaft and provide feedback to control circuitry, and control circuitry **68** for receiving control signals and actuating the drive unit. Each drive unit **63** being independently controlled and motorized, the instrument driver **62** may provide multiple (e.g., four as shown in FIG. **15**) independent drive outputs to the medical instrument. In operation, the control circuitry **68** would receive a control signal, transmit a motor signal to the motor **66**, compare the resulting motor speed as measured by the encoder **67** with the desired speed, and modulate the motor signal to generate the desired torque.

[0134] For procedures that require a sterile environment, the robotic system may incorporate a drive interface, such as a sterile adapter connected to a sterile drape, that sits between the instrument driver and the medical instrument. The chief purpose of the sterile adapter is to transfer angular motion from the drive shafts of the instrument driver to the drive inputs of the instrument while maintaining physical separation, and thus sterility, between the drive shafts and drive inputs. Accordingly, an example sterile adapter may comprise a series of rotational inputs and outputs intended to be mated with the drive shafts of the instrument driver and drive inputs on the instrument. Connected to the sterile adapter, the sterile drape, comprised of a thin, flexible material such as transparent or translucent plastic, is designed to cover the capital equipment, such as the instrument driver, robotic arm, and cart (in a cart-based system) or table (in a table-based system). Use of the drape would allow the capital equipment to be positioned proximate to the patient while still being located in an area not requiring sterilization (i.e., non-sterile field). On the other side of the sterile drape, the medical instrument may interface with the patient in an area requiring sterilization (i.e., sterile field).

D. Medical Instrument.

[0135] FIG. **16** illustrates an example medical instrument with a paired instrument driver. Like other instruments designed for use with a robotic system, medical instrument **70** comprises an elongated shaft **71** (or elongate body) and an instrument base **72**. The instrument base **72**, also referred to as an “instrument handle” due to its intended design for manual interaction by the physician, may generally comprise rotatable drive inputs **73**, e.g., receptacles, pulleys or spools, that are designed to be mated with drive outputs **74** that extend through a drive interface on instrument driver **75** at the distal end of robotic arm **76**. When physically connected, latched, and/or coupled, the mated drive inputs **73** of the instrument base **72** may share axes of rotation with the drive outputs **74** in the instrument driver **75** to allow the transfer of torque from the drive outputs **74** to the drive inputs **73**. In some embodiments, the drive outputs **74** may comprise splines that are designed to mate with receptacles on the drive inputs **73**.

[0136] The elongated shaft **71** is designed to be delivered through either an anatomical opening or lumen, e.g., as in endoscopy, or a minimally invasive incision, e.g., as in laparoscopy. The elongated shaft **71** may be either flexible (e.g., having properties similar to an endoscope) or rigid (e.g., having properties similar to a laparoscope) or contain a customized combination of both flexible and rigid portions. When designed for laparoscopy, the distal end of a rigid elongated shaft may be connected to an end effector extending from a jointed wrist formed from a clevis with at least one degree of freedom and a surgical tool or medical instrument, such as, for example, a grasper or scissors, that may be actuated based on force from the tendons as the drive inputs rotate in response to torque received from the drive outputs **74** of the instrument driver **75**. When designed for endoscopy, the distal end of a flexible elongated shaft may include a steerable or controllable bending section that may be articulated and bent based on torque received from the drive outputs **74** of the instrument driver **75**.

[0137] Torque from the instrument driver **75** is transmitted down the elongated shaft **71** using tendons along the elongated shaft **71**. These individual tendons, such as pull wires, may be individually anchored to individual drive inputs **73** within the instrument handle **72**. From the

instrument handle **72**, the tendons are directed down one or more pull lumens along the elongated shaft **71** and anchored at the distal portion of the elongated shaft **71**, or in the wrist at the distal portion of the elongated shaft. During a surgical procedure, such as a laparoscopic, endoscopic or hybrid procedure, these tendons may be coupled to a distally mounted end effector, such as a wrist, grasper, or scissor. Under such an arrangement, torque exerted on drive inputs **73** would transfer tension to the tendon, thereby causing the end effector to actuate in some way. In some embodiments, during a surgical procedure, the tendon may cause a joint to rotate about an axis, thereby causing the end effector to move in one direction or another. Alternatively, the tendon may be connected to one or more jaws of a grasper at the distal end of the elongated shaft **71**, where tension from the tendon causes the grasper to close.

[0138] In endoscopy, the tendons may be coupled to a bending or articulating section positioned along the elongated shaft **71** (e.g., at the distal end) via adhesive, control ring, or other mechanical fixation. When fixedly attached to the distal end of a bending section, torque exerted on the drive inputs **73** would be transmitted down the tendons, causing the softer, bending section (sometimes referred to as the articulable section or region) to bend or articulate. Along the non-bending sections, it may be advantageous to spiral or helix the individual pull lumens that direct the individual tendons along (or inside) the walls of the endoscope shaft to balance the radial forces that result from tension in the pull wires. The angle of the spiraling and/or spacing therebetween may be altered or engineered for specific purposes, wherein tighter spiraling exhibits lesser shaft compression under load forces, while lower amounts of spiraling results in greater shaft compression under load forces, but limits bending. On the other end of the spectrum, the pull lumens may be directed parallel to the longitudinal axis of the elongated shaft **71** to allow for controlled articulation in the desired bending or articulable sections.

[0139] In endoscopy, the elongated shaft **71** houses a number of components to assist with the robotic procedure. The shaft **71** may comprise a working channel for deploying surgical tools (or medical instruments), irrigation, and/or aspiration to the operative region at the distal end of the shaft **71**. The shaft **71** may also accommodate wires and/or optical fibers to transfer signals to/from an optical assembly at the distal tip, which may include an optical camera. The shaft **71** may also accommodate optical fibers to carry light from proximally-located light sources, such as light emitting diodes, to the distal end of the shaft **71**.

[0140] At the distal end of the instrument **70**, the distal tip may also comprise the opening of a working channel for delivering tools for diagnostic and/or therapy, irrigation, and aspiration to an operative site. The distal tip may also include a port for a camera, such as a fiberscope or a digital camera, to capture images of an internal anatomical space. Relatedly, the distal tip may also include ports for light sources for illuminating the anatomical space when using the camera.

[0141] In the example of FIG. **16**, the drive shaft axes, and thus the drive input axes, are orthogonal to the axis of the elongated shaft **71**. This arrangement, however, complicates roll capabilities for the elongated shaft **71**. Rolling the elongated shaft **71** along its axis while keeping the drive inputs **73** static results in undesirable tangling of the tendons as they extend off the drive inputs **73** and enter pull lumens within the elongated shaft **71**. The resulting entanglement of such tendons may disrupt any control algorithms intended to predict movement of the flexible elongated shaft **71** during an endoscopic procedure.

[0142] FIG. **17** illustrates an alternative design for an instrument driver and instrument where the axes of the drive units are parallel to the axis of the elongated shaft of the instrument. As shown, a circular instrument driver **80** comprises four drive units with their drive outputs **81** aligned in parallel at the end of a robotic arm **82**. The drive units, and their respective drive outputs **81**, are housed in a rotational assembly **83** of the instrument driver **80** that is driven by one of the drive units within the assembly **83**. In response to torque provided by the rotational drive unit, the rotational assembly **83** rotates along a circular bearing that connects the rotational assembly **83** to the non-rotational portion **84** of the instrument driver **80**. Power and controls signals may be

communicated from the non-rotational portion **84** of the instrument driver **80** to the rotational assembly **83** through electrical contacts that may be maintained through rotation by a brushed slip ring connection (not shown). In other embodiments, the rotational assembly **83** may be responsive to a separate drive unit that is integrated into the non-rotatable portion **84**, and thus not in parallel to the other drive units. The rotational mechanism **83** allows the instrument driver **80** to rotate the drive units, and their respective drive outputs **81**, as a single unit around an instrument driver axis **85**.

[0143] Like earlier disclosed embodiments, an instrument **86** may comprise an elongated shaft portion **88** and an instrument base **87** (shown with a transparent external skin for discussion purposes) comprising a plurality of drive inputs **89** (such as receptacles, pulleys, and spools) that are configured to receive the drive outputs **81** in the instrument driver **80**. Unlike prior disclosed embodiments, the instrument shaft **88** extends from the center of the instrument base **87** with an axis substantially parallel to the axes of the drive inputs **89**, rather than orthogonal as in the design of FIG. **16**.

[0144] When coupled to the rotational assembly **83** of the instrument driver **80**, the medical instrument **86**, comprising instrument base **87** and instrument shaft **88**, rotates in combination with the rotational assembly **83** about the instrument driver axis **85**. Since the instrument shaft **88** is positioned at the center of instrument base **87**, the instrument shaft **88** is coaxial with instrument driver axis **85** when attached. Thus, rotation of the rotational assembly **83** causes the instrument shaft **88** to rotate about its own longitudinal axis. Moreover, as the instrument base **87** rotates with the instrument shaft **88**, any tendons connected to the drive inputs **89** in the instrument base **87** are not tangled during rotation. Accordingly, the parallelism of the axes of the drive outputs **81**, drive inputs **89**, and instrument shaft **88** allows for the shaft rotation without tangling any control tendons.

[0145] FIG. **18** illustrates an instrument having an instrument based insertion architecture in accordance with some embodiments. The instrument **150** can be coupled to any of the instrument drivers discussed above. The instrument **150** comprises an elongated shaft **152**, an end effector **162** connected to the shaft **152**, and a handle **170** coupled to the shaft **152**. The elongated shaft **152** comprises a tubular member having a proximal portion **154** and a distal portion **156**. The elongated shaft **152** comprises one or more channels or grooves **158** along its outer surface. The grooves **158** are configured to receive one or more wires or cables **180** therethrough. One or more cables **180** thus run along an outer surface of the elongated shaft **152**. In other embodiments, cables **180** can also run through the elongated shaft **152**. Manipulation of the one or more cables **180** (e.g., via an instrument driver) results in actuation of the end effector **162**.

[0146] The instrument handle **170**, which may also be referred to as an instrument base, may generally comprise an attachment interface **172** having one or more mechanical inputs **174**, e.g., receptacles, pulleys or spools, that are designed to be reciprocally mated with one or more torque couplers on an attachment surface of an instrument driver.

[0147] In some embodiments, the instrument **150** comprises a series of pulleys or cables that enable the elongated shaft **152** to translate relative to the handle **170**. In other words, the instrument **150** itself comprises an instrument-based insertion architecture that accommodates insertion of the instrument, thereby minimizing the reliance on a robot arm to provide insertion of the instrument **150**. In other embodiments, a robotic arm can be largely responsible for instrument insertion.

E. Controller.

[0148] Any of the robotic systems described herein can include an input device or controller for manipulating an instrument attached to a robotic arm. In some embodiments, the controller can be coupled (e.g., communicatively, electronically, electrically, wirelessly and/or mechanically) with an instrument such that manipulation of the controller causes a corresponding manipulation of the instrument e.g., via master slave control.

[0149] FIG. **19** is a perspective view of an embodiment of a controller **182**. In the present

embodiment, the controller **182** comprises a hybrid controller that can have both impedance and admittance control. In other embodiments, the controller **182** can utilize just impedance or passive control. In other embodiments, the controller **182** can utilize just admittance control. By being a hybrid controller, the controller **182** advantageously can have a lower perceived inertia while in use.

[0150] In the illustrated embodiment, the controller **182** is configured to allow manipulation of two medical instruments, and includes two handles **184**. Each of the handles **184** is connected to a gimbal **186**. Each gimbal **186** is connected to a positioning platform **188**.

[0151] As shown in FIG. **19**, each positioning platform **188** includes a selective compliance assembly robot arm (SCARA) **198** coupled to a column **194** by a prismatic joint **196**. The prismatic joints **196** are configured to translate along the column **194** (e.g., along rails **197**) to allow each of the handles **184** to be translated in the z-direction, providing a first degree of freedom. The SCARA **198** is configured to allow motion of the handle **184** in an x-y plane, providing two additional degrees of freedom.

[0152] In some embodiments, one or more load cells are positioned in the controller. For example, in some embodiments, a load cell (not shown) is positioned in the body of each of the gimbals **186**. By providing a load cell, portions of the controller **182** are capable of operating under admittance control, thereby advantageously reducing the perceived inertia of the controller while in use. In some embodiments, the positioning platform **188** is configured for admittance control, while the gimbal **186** is configured for impedance control. In other embodiments, the gimbal **186** is configured for admittance control, while the positioning platform **188** is configured for impedance control. Accordingly, for some embodiments, the translational or positional degrees of freedom of the positioning platform **188** can rely on admittance control, while the rotational degrees of freedom of the gimbal **186** rely on impedance control.

F. Navigation and Control.

[0153] Traditional endoscopy may involve the use of fluoroscopy (e.g., as may be delivered through a C-arm) and other forms of radiation-based imaging modalities to provide endoluminal guidance to an operator physician. In contrast, the robotic systems contemplated by this disclosure can provide for non-radiation-based navigational and localization means to reduce physician exposure to radiation and reduce the amount of equipment within the operating room. As used herein, the term “localization” may refer to determining and/or monitoring the position of objects in a reference coordinate system. Technologies such as preoperative mapping, computer vision, real-time EM tracking, and robot command data may be used individually or in combination to achieve a radiation-free operating environment. In other cases, where radiation-based imaging modalities are still used, the preoperative mapping, computer vision, real-time EM tracking, and robot command data may be used individually or in combination to improve upon the information obtained solely through radiation-based imaging modalities.

[0154] FIG. **20** is a block diagram illustrating a localization system **90** that estimates a location of one or more elements of the robotic system, such as the location of the instrument, in accordance to an example embodiment. The localization system **90** may be a set of one or more computer devices configured to execute one or more instructions. The computer devices may be embodied by a processor (or processors) and computer-readable memory in one or more components discussed above. By way of example and not limitation, the computer devices may be in the tower **30** shown in FIG. **1**, the cart **11** shown in FIGS. **1-4**, the beds shown in FIGS. **5-14**, etc.

[0155] As shown in FIG. **20**, the localization system **90** may include a localization module **95** that processes input data **91-94** to generate location data **96** for the distal tip of a medical instrument. The location data **96** may be data or logic that represents a location and/or orientation of the distal end of the instrument relative to a frame of reference. The frame of reference can be a frame of reference relative to the anatomy of the patient or to a known object, such as an EM field generator (see discussion below for the EM field generator).

[0156] The various input data **91-94** are now described in greater detail. Preoperative mapping may be used by the localization module **95** to generate model data **91**. Preoperative mapping may be accomplished through the use of the collection of low dose CT scans. Preoperative CT scans are reconstructed into three-dimensional images, which are visualized, e.g. as “slices” of a cutaway view of the patient's internal anatomy. When analyzed in the aggregate, image-based models for anatomical cavities, spaces and structures of the patient's anatomy, such as a patient lung network, may be generated. Techniques such as center-line geometry may be determined and approximated from the CT images to develop a three-dimensional volume of the patient's anatomy, referred to as model data **91** (also referred to as “preoperative model data” when generated using only preoperative CT scans). The use of center-line geometry is discussed in U.S. patent application Ser. No. 14/523,760, the contents of which are herein incorporated in its entirety. Network topological models may also be derived from the CT-images, and are particularly appropriate for bronchoscopy.

[0157] In some embodiments, the instrument may be equipped with a camera to provide vision data (or image data) **92** to the localization module **95**. The localization module **95** may process the vision data **92** to enable one or more vision-based (or image-based) location tracking modules or features. For example, the preoperative model data **91** may be used in conjunction with the vision data **92** to enable computer vision-based tracking of the medical instrument (e.g., an endoscope or an instrument advance through a working channel of the endoscope). For example, using the preoperative model data **91**, the robotic system may generate a library of expected endoscopic images from the model based on the expected path of travel of the endoscope, each image linked to a location within the model. Intraoperatively, this library may be referenced by the robotic system in order to compare real-time images captured at the camera (e.g., a camera at a distal end of the endoscope) to those in the image library to assist localization.

[0158] Other computer vision-based tracking techniques use feature tracking to determine motion of the camera, and thus the endoscope. Some features of the localization module **95** may identify circular geometries in the preoperative model data **91** that correspond to anatomical lumens and track the change of those geometries to determine which anatomical lumen was selected, as well as the relative rotational and/or translational motion of the camera. Use of a topological map may further enhance vision-based algorithms or techniques.

[0159] Optical flow, another computer vision-based technique, may analyze the displacement and translation of image pixels in a video sequence in the vision data **92** to infer camera movement. Examples of optical flow techniques may include motion detection, object segmentation calculations, luminance, motion compensated encoding, stereo disparity measurement, etc. Through the comparison of multiple frames over multiple iterations, movement and location of the camera (and thus the endoscope) may be determined.

[0160] The localization module **95** may use real-time EM tracking and EM data **93** to generate a real-time location of the endoscope in a global coordinate system that may be registered to the patient's anatomy, represented by the preoperative model. In EM tracking, an EM sensor (or tracker) comprising one or more sensor coils embedded in one or more locations and orientations in a medical instrument (e.g., an endoscopic tool) measures the variation in the EM field created by one or more static EM field generators positioned at a known location. The location information detected by the EM sensors is stored as EM data **93**. The EM field generator (or transmitter), may be placed close to the patient to create a low intensity magnetic field that the embedded sensor may detect. The magnetic field induces small currents in the sensor coils of the EM sensor, which may be analyzed to determine the distance and angle between the EM sensor and the EM field generator. These distances and orientations may be intraoperatively “registered” to the patient anatomy (e.g., the preoperative model) in order to determine the geometric transformation that aligns a single location in the coordinate system with a position in the preoperative model of the patient's anatomy. Once registered, an embedded EM tracker in one or more positions of the

medical instrument (e.g., the distal tip of an endoscope) may provide real-time indications of the progression of the medical instrument through the patient's anatomy.

[0161] Robotic command and kinematics data **94** may also be used by the localization module **95** to provide location data **96** for the robotic system. Device pitch and yaw resulting from articulation commands may be determined during preoperative calibration. Intraoperatively, these calibration measurements may be used in combination with known insertion depth information to estimate the position of the instrument. Alternatively, these calculations may be analyzed in combination with EM, vision, and/or topological modeling to estimate the position of the medical instrument within the network.

[0162] As FIG. **20** shows, a number of other input data can be used by the localization module **95**. For example, although not shown in FIG. **20**, an instrument utilizing shape-sensing fiber can provide shape data that the localization module **95** can use to determine the location and shape of the instrument.

[0163] The localization module **95** may use the input data **91-94** in combination(s). In some cases, such a combination may use a probabilistic approach where the localization module **95** assigns a confidence weight to the location determined from each of the input data **91-94**. Thus, where the EM data may not be reliable (as may be the case where there is EM interference) the confidence of the location determined by the EM data **93** can be decrease and the localization module **95** may rely more heavily on the vision data **92** and/or the robotic command and kinematics data **94**.

[0164] As discussed above, the robotic systems discussed herein may be designed to incorporate a combination of one or more of the technologies above. The robotic system's computer-based control system, based in the tower, bed and/or cart, may store computer program instructions, for example, within a non-transitory computer-readable storage medium such as a persistent magnetic storage drive, solid state drive, or the like, that, upon execution, cause the system to receive and analyze sensor data and user commands, generate control signals throughout the system, and display the navigational and localization data, such as the position of the instrument within the global coordinate system, anatomical map, etc.

2. Introduction to Detection of Contact Between a Link and an External Object

[0165] Embodiments of the disclosure relate to systems and techniques for detecting contact between a robotic arm/manipulator and an external object. As described herein, robotic medical systems may include a plurality of robotic arms configured to control the movement of medical tool(s) during a given medical procedure. In order to achieve a desired pose of a medical tool, a robotic arm may be placed into a pose, which may cause the robotic arm to come into contact with an external object, such as, for example, a patient, bedside staff, or inanimate object(s) (e.g., accessories on the bed). If undetected, such contact (also referred to as a "collision") can result in injury to the patient or bedside staff and/or damage to the robotic arm or external object(s). By detecting contact between the robotic arm and the external object early enough (e.g., before the contact results in a force greater than a threshold value), injury and/or damage can be reduced or prevented.

[0166] One safety consideration relates to when a robotic link (e.g., a portion of robotic arm) comes into contact with an external object. For example, when the robotic arm comes into contact with a patient in an unplanned or accidental manner, harmful forces may be created and/or transferred to the patient. Thus, one challenge with robotic medical systems is detecting contact with surrounding objects and people (including the patient) to ensure that the robotic system can safely operate around people.

[0167] Aspects of this disclosure provide systems and techniques for detecting contact between or with links (e.g., of the robotic arms), which can involve the use of one or more sensors. Once contact is detected, the robotic system can take action to reduce further contact. For example, in some implementations the robotic arm can stop further movement upon detection of contact with an external object.

Contact Detection Using a Force Sensing Floating Shell

[0168] According to aspects of this disclosure, a robotic arm can comprise a plurality of robotic links connected via a series of joints providing a plurality of degrees-of-freedom of movement that enable the robotic arm to be positioned in a variety of different poses or states. FIGS. **21A** and **21B** illustrate example robotic links that can form a portion of a robotic arm in accordance with aspects of this disclosure. Although examples of robotic links used as a part of a robotic arm are provided herein, aspects of this disclosure also include other types of links (e.g., robotic and non-robotic, medical and non-medical) for which detection of contact with external objects may be desirable.

[0169] FIG. **21A** illustrates three views of a distal portion of a robotic arm **201**, whereas FIG. **21B** illustrates three views of a portion of the robotic arm **201** that is proximal to the distal portion of FIG. **21A**. With reference to FIGS. **21A** and **21B**, the robotic arm **201** may include a device manipulator **203**, a plurality of links **205**, **207**, **209**, and **211**, and a plurality of joints **213**, **215**, **217**, and **219** connecting the device manipulator **203** and the links **205-211**. In each of the views illustrated in FIG. **21A**, regions **221** of the robotic arm **201** which have a relatively higher likelihood of colliding with a patient are highlighted.

[0170] Aspects of this disclosure are directed to an arm component (e.g., one of the robotic links **205-211** or joints **213-219** of FIGS. **21A-21B**) that is capable of sensing contact with external objects such as a patient, bedside staff, or other object.

[0171] In certain implementations, a shell can be suspended around a given link and the relative motion between the shell and inner components/members of the link can be detected using one or more sensors in order to detect contact with an external object. FIG. **22** illustrates an example link **300** including a rigid shell **309** and configured to detect contact with external objects in accordance with aspects of this disclosure. In particular, the link **300** includes a structural link **301**, a structural cover **303**, a first joint **305**, a second joint **307**, a shell **309**, a pair of reaction paddles **311**, and a shell cover **313**. For example, the inner components of the link **300** may comprise the structural link **301** and the structural cover **303**.

[0172] The structural cover **303** can be attached to the structural link **301** to house components of the structural link **301** and form the internal structural connection between the first joint **305** and the second link **305**. The shell **309** together with the shell cover **313** are suspended from and surround the structural link **301**. As used herein, the shell **309** and shell cover **313** may collectively be referred to simply as the “shell” **309**, while the structural link **301** and structural cover **303** may collectively be referred to simply as the structural link **301** or a manipulatable link, unless the context clearly indicates otherwise.

[0173] The shell **309** may be connected to the structural link **301** via a force-sensing connection. Because the shell **309** surrounds the structural link **301**, when the link **300** contacts an external object, the object will come into contact with the shell **309**. Thus, the force-sensing connection can detect contact between the shell **309** and the external object by measuring changes in the force between the shell **309** and the structural link **301** caused by the link **300** coming into contact with the external object. The shell **309** may also be sufficiently rigid such that, upon contact with an external object, the shell **309** engages the force-sensing connection. Advantageously, by using a rigid shell **309** the force and relative movement between the shell **309** and the structural link **301** can be sensed in all three directions.

[0174] The force-sensing connection may be implemented in a variety of different manners in accordance with aspects of this disclosure. For example, the force-sensing connection can include one or more of the following: traditional load cells, force sensing resistors, and/or any component capable of sensing force (or displacement, when combined with a spring).

[0175] FIG. **23** illustrates one example of a force-sensing connection that can be used in the link **300** of FIG. **22** in accordance with aspects of this disclosure. In particular, the force-sensing connection can include a plurality of shell sensors **321** (e.g., fourteen shell sensors in the illustrated embodiment), which can be disposed between the structural link **301** and the shell **309**.

[0176] In some embodiments, the sensors **321** are distributed throughout the link **300** in between the shell **309** and the structural link **301**. For example, the shell **309** can be suspended over the structural link **301** via the sensors **321**. Depending on the implementation, the link **300** can include one, two, three, four or more sensors **321** distributed along a robotic arm link. FIG. **24A** illustrates two views of an embodiment in which fourteen sensors **321** are included in a link **401** in accordance with aspects of this disclosure. In particular, FIG. **24A** illustrates a side and front view of one end of the link **401** which includes seven sensors **321**. The link **401** may be substantially symmetric at both ends of the link **401**, thereby including a total of fourteen sensors **321** in the link **401**.

[0177] FIG. **24B** illustrates two view of an embodiment in which twelve sensors **321** are included in a link **411** in accordance with aspects of this disclosure. In some implementations, a robotic arm may include the link **401** and the link **411**, with the link **401** positioned proximal to the link **411**. In particular, FIG. **24B** illustrates a side and front view of the link **411** which includes twelve sensors **321**. The link **401** may be substantially symmetric at both ends of the link **401**, thereby including a total of fourteen sensors **321** in the link **401**. In a number of implementations, the plurality of sensors **321** can be configured to support the rigid shell without being fixed to the rigid shell. In some implementations, the link **401**, **411** may further include one or more supports configured to support the rigid shell relative to the structural link. For example, the one or more supports can include spring(s), flexure(s), and/or suspension(s).

[0178] Although FIGS. **24A** and **24B** illustrate embodiments of links **401**, **411** including a plurality of sensors **321**, in other implementations the link can include a single sensor configured to sense force and/or displacement between the structural link **301** and the shell **309** in multiple directions. Using signals received from the sensor(s) **321**, the robotic system can be configured to detect a direction of the contact between the shell **309** and the external object. The robotic system can also measure a magnitude of a force resulting from the contact between the shell **309** and the external object based on the signal from the sensor(s) **321**. Based on the placement of the plurality of sensors **321** within the links **401** and **411**, the robotic system can also be configured to detect a torque applied to the link. For example, if a torque is applied to the shell **309**, certain sensors **321** on one side of the link **401**, **411** may be compressed. Based on the positions of and forces sensed by the sensors **321** being compressed, the robotic system can determine a torque applied to the link **401**, **411**.

[0179] FIG. **25** is a flowchart illustrating an example method **500** operable by a surgical robotic system, or component(s) thereof, for detecting contact between a robotic arm/manipulator and an external object in accordance with one or more aspects of this disclosure. It is to be appreciated that the steps of method **500** illustrated in FIG. **25** may be performed by one or more processors of a surgical robotic system. For convenience, the method **500** is described as performed by a processor of the system.

[0180] The processor may be included as a part of a system, including a robotic manipulator including a link, a rigid shell configured to overlay the link, and one or more sensors positioned between the rigid shell and the link. The one or more sensors are configured to detect contact between one or more external objects and the rigid shell in at least two or more directions. The system may further include at least one computer-readable memory in communication with the processor and having stored thereon computer-executable instructions to cause the set of processors to perform the method **500**.

[0181] The method **500** may begin at block **501**. At block **502**, the processor may receive a signal from the one or more sensors positioned between the manipulatable link of the robotic manipulator and the rigid shell configured to overlay the link. The received signal is indicative of the degree of contact between the rigid shell and the one or more sensors.

[0182] At block **504**, the processor may determine that contact between the robotic manipulator and the external object has occurred in response to the received signal meeting or exceeding a threshold

value. In some implementations, the processor can also detect a direction of the contact between the rigid shell and the external object based on the received signal. The processor can also measure a magnitude of a force and/or torque resulting from the contact between the rigid shell and the external object based on information in the signal (e.g., data in the time or frequency domain) received from the one or more sensors. The processor may also be configured to measure a direction of relative motion between the rigid shell and the link based on the received signal. [0183] In certain embodiments, the processor may also be configured to determine the location in space of the external object based on signal(s) from sensor(s) indicative of the detected contact and prevent movement of the robotic arm (or other robotic arms) to the location of the contact by updating a collision model to avoid future contact in the same area. Such information may include which sensors are activated by the contact and/or directional information detected by a given sensor. As used herein, a collision model may be a model of object(s) within the environment which may be at risk of a collision with the robotic arms. The robotic system can prevent movement of the robotic arms which would result in collisions with object(s) modeled in the collision model. The method **500** may end at block **506**.

[0184] The one or more sensors **321** that are provided in the link **300**, **401**, or **411** can include: traditional load cells, force sensing resistors, strain gauges, and/or capacitive sensors, etc. In one implementation, the one or more sensors **321** comprise beam flexure type sensors **600** (also referred to a spring or diving board sensor). FIG. **26** illustrates an embodiment of a beam flexure sensor **600** in accordance with aspects of this disclosure. In this implementation, the sensor **600** comprises a beam flexure **601** configured to contact an underlying force sensor **603**. Upon a force being applied to the shell **309**, the shell applies a force to the sensor through the beam flexure **601** which then applies a force to the force sensor **603** (which may be scaled from the original force). Such motions (which can include micromotions) will therefore allow the sensor **603** to detect that contact with an external object has occurred. Based on the detection of contact with an external object, the robotic system can take actions (e.g., stop motion of the robotic arm) to reduce the contact or prevent further contact.

[0185] FIG. **27** illustrates another embodiment of a beam flexure sensor **700** in accordance with aspects of this disclosure. In this implementation, the sensor **700** comprises a beam flexure **701** that is configured to contact an underlying force sensor **703**, and a contact point **705** at which the beam flexure **701** is configured to contact the shell **309**. Although not illustrated, the sensor **700** can further include a circuit board configured to receive an output signal from the force sensor **703** and output a signal indicative of the force applied to the contact point **705**. In some implementations, the force sensor **703** may be mounted on the circuit board. The beam flexure **701** may comprise a first end (e.g., on the left side of FIG. **27**) that is fixed and a second end (e.g., on the right side of FIG. **27**) coupled to the force sensor **703**. The contact point **705** is located between the first and second ends of the beam flexure **701**. Accordingly, the force exerted on the force sensor **703** is proportional to or otherwise indicative of the force applied to the contact point **705**. In combination with the contact point **705**, the beam flexure **701** can be configured to scale a force applied to the sensor **700** and isolate the sensor **703** from shear forces applied to the shell **309**. In certain implementations, the contact point **705** may function as a load concentrator configured to control a point at which the shell **309** applies a force to the sensor **700** (e.g., the point at which the force is applied to the beam flexure **701**).

[0186] The specific location of the contact point **705** between the first and second ends will affect the amount of leverage provided by contact of an external object with the shell **309** onto the force sensor **703**. For example, when the contact point **705** is located closer to the second end with the force sensor **703**, more of the force applied to the contact point **705** will be transmitted to the force sensor **703** than when the contact point **705** is located closer to the fixed first end. Thus, the location of the contact point **705** can be selected to map the range of forces expected to be received by the shell **309** onto the range of forces detectable by the force sensor **703**. By reducing the forces

using the beam flexure **701**, the force sensor **703** can effectively measure greater forces applied to the shell **309** by reducing the received force onto the range of forces measurable by the force sensor **703**.

[0187] Aspects of this disclosure provide a number of advantages for the detection of contact between a link and an external object. For example, aspects of this disclosure are advantageous over systems that measure torque (either directly with a torque sensor, or through current or control loop disturbances) as the sensitivity of implementation of this disclosure do not decrease as the applied force vector approaches the torque axis, as is the case in torque sensing systems. Similarly, systems that mount the entire robot on load cells (as is common in some industrial robots) may have similar drawbacks to torque-based measurement systems.

[0188] Additionally, by using a shell **309** to detect contact with an external object as described herein, the system can advantageously sense contact on the entire surface of the link. In contrast, implementations which employ a film to sense contact may not be able to sense or detect contact with an external object at locations where the film is not applied to the link(s).

[0189] Another advantage of aspects of this disclosure includes allowing for the system to sense the total amount of contact force over the entire surface of a given link, to allow a certain amount of light incidental contact without interrupting a medical procedure. This is advantageous over systems that are not able to distinguish between a small amount of incidental force and larger forces that risk injury and/or damage.

[0190] Yet another advantage of aspects of this disclosure is the compact form factor enabled by the shell compared to systems that surround the robot in pressure-sensitive gas-filled bladders.

Contact Detection Using a Film

[0191] In certain implementations, a coating or film of sensors can be adhered to a surface of a link (e.g., a robotic arm link). FIG. **28** illustrates an example robotic link **800** that can form a portion of a robotic arm in accordance with aspects of this disclosure. In particular, the link **800** includes a structural link **801** and a film **803** applied to an external surface of the structural link **801**.

Depending on the implementation, the film **803** can comprise various types of force sensors. For example, technologies which can be used in the film **803** include: force sensing resistors (FSR), capacitive sensors, proximity sensors, and/or others.

[0192] In certain implementations, the film **803** may be attached to surfaces of the link **800** that are more likely to contact external objects (e.g., the regions **221** illustrated in FIGS. **21A** and **21B**). By detecting contact with external objects using the film **803**, the robotic system can respond to the detected contact to prevent applying an unacceptably high load onto the external object (e.g., an inanimate object such as medical accessories or a bed, beside staff, or the patient).

[0193] By applying such a film **803** to a robotic arm, the robotic system can detect contact between an external object and the robotic arm and prevent further movement in the direction of the contact with the external object. In addition, the robotic system can also determine the location in space of the external object based on the detected contact and prevent movement of the robotic arm (or other robotic arms) to the location of the contact by updating a collision model to avoid future contact in the same area. The robotic system can prevent movement of the robotic arms which would result in collisions with object(s) modeled in the collision model.

[0194] The use of a force sensing film **803** in accordance with aspects of this disclosure allows for the maximization of a robot's collision free workspace by allowing the robot (e.g., a robotic arm) to move until it collides with an external object and then stop. This allows for a larger workspace than might otherwise be feasible with force sensing solutions that occupy a larger volume around the structural link (e.g., systems that surround the robot in pressure-sensitive gas-filled bladders).

[0195] In contrast to systems that use force or torque sensors positioned in robotic arm joints, aspects of this disclosure which employ force sensing films do not experience a decrease as the applied force vector approaches the torque axis, as is the case in torque sensing systems. Similarly, systems that mount the entire robot on load cells (as is common in some industrial robots) may

have similar drawbacks to torque-based measurement systems.

[0196] The advantage of an increased collision-free workspace is also apparent compared to systems which artificially limiting a robot's workspace such that it can never contact the patient or accessories. Since such artificial limitations cannot precisely define the locations of external objects, implementations which use such artificial limitations can dramatically limit the workspace and cause user irritation due to unachievable movements and/or poses of the robotic arms. In related aspects, force sensing film **803** may be used in conjunction with one or more sensors **321** that are provided in a manipulatable link, wherein the one or more sensors can include: traditional load cells, force sensing resistors, strain gauges, and/or capacitive sensors, etc.

4. Implementing Systems and Terminology

[0197] Implementations disclosed herein provide systems, methods and apparatus for detecting contact between a link and an external object.

[0198] It should be noted that the terms “couple,” “coupling,” “coupled” or other variations of the word couple as used herein may indicate either an indirect connection or a direct connection. For example, if a first component is “coupled” to a second component, the first component may be either indirectly connected to the second component via another component or directly connected to the second component.

[0199] The detection of contact between a link and an external object functions described herein may be stored as one or more instructions on a processor-readable or computer-readable medium. The term “computer-readable medium” refers to any available medium that can be accessed by a computer or processor. By way of example, and not limitation, such a medium may comprise random access memory (RAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), flash memory, compact disc read-only memory (CD-ROM) or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to store desired program code in the form of instructions or data structures and that can be accessed by a computer. It should be noted that a computer-readable medium may be tangible and non-transitory. As used herein, the term “code” may refer to software, instructions, code or data that is/are executable by a computing device or processor.

[0200] The methods disclosed herein comprise one or more steps or actions for achieving the described method. The method steps and/or actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps or actions is required for proper operation of the method that is being described, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims.

[0201] As used herein, the term “plurality” denotes two or more. For example, a plurality of components indicates two or more components. The term “determining” encompasses a wide variety of actions and, therefore, “determining” can include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), ascertaining and the like. Also, “determining” can include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Also, “determining” can include resolving, selecting, choosing, establishing and the like.

[0202] The phrase “based on” does not mean “based only on,” unless expressly specified otherwise. In other words, the phrase “based on” describes both “based only on” and “based at least on.”

[0203] The previous description of the disclosed implementations is provided to enable any person skilled in the art to make or use the present invention. Various modifications to these implementations will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other implementations without departing from the scope of the invention. For example, it will be appreciated that one of ordinary skill in the art will be able to employ a number corresponding alternative and equivalent structural details, such as equivalent ways of fastening, mounting, coupling, or engaging tool components, equivalent mechanisms for

producing particular actuation motions, and equivalent mechanisms for delivering electrical energy. Thus, the present invention is not intended to be limited to the implementations shown herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

Claims

1-20. (canceled)

21. A method comprising: receiving, by one or more processors, a signal from one or more sensors positioned between a link of a robotic manipulator and a rigid shell formed over the link; and determining, by the one or more processors, that contact between the robotic manipulator and an external object has occurred in response to the received signal meeting or exceeding a threshold value.

22. The method of claim 21, the external object comprising one or more of: a patient, a clinician, or an inanimate object.

23. The method of claim 21, the one or more sensors being positioned in an interstitial space between the link and the rigid shell.

24. The method of claim 21, the one or more sensors measuring a magnitude of a force applied to the link.

25. The method of claim 21, the one or more sensors measuring a direction of a force applied to the link.

26. The method of claim 21, the one or more sensors measuring a torque applied to the link.

27. The method of claim 21, the one or more sensors measuring a relative position of the rigid shell relative to the link.

28. The method of claim 21, the robotic manipulator comprising a support that supports the rigid shell relative to the link, the support comprising at least one of a spring, a flexure, or a suspension.

29. The method of claim 21, the one or more sensors measuring a direction of relative motion between the rigid shell and the link.

30. The method of claim 21, the one or more sensors comprising a plurality of sensors configured to detect contact between the external object and the rigid shell, the plurality of sensors being positioned at different locations of the link.

31. The method of claim 21, the one or more sensors comprising one or more circuit board mounted force sensors.

32. The method of claim 21, the one or more sensors comprising one or more strain gauges.

33. The method of claim 21, the one or more sensors comprising one or more force sensing resistors.

34. The method of claim 21, the one or more sensors comprising one or more capacitive sensors.

35. The method of claim 21, further comprising controlling, using a load concentrator, a point at which the rigid shell applies force to the one or more sensors.

36. The method of claim 21, further comprising scaling, using a flexure, a force applied to the one or more sensors and isolating the one or more sensors from shear forces.

37. The method of claim 21, further comprising updating a collision model based on the received signal.

38. The method of claim 21, further comprising determining a location in space of the external object based on the received signal.

39. A method comprising: receiving, by one or more processors, a signal from one or more sensors positioned between a link of a robotic manipulator and a rigid shell formed over the link; and determining, by the one or more processors, that contact between the robotic manipulator and an external object has occurred based on the received signal.

40. The method of claim 39, further comprising: updating a collision model based on the received

signal; and controlling the robotic manipulator based on the updated collision model to avoid future contact with the external object.
